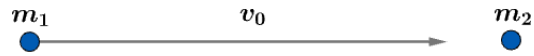


2008 F=ma Exam: Problem 9

Kevin S. Huang



By conservation of linear momentum, we have

$$m_1 \vec{v}_0 = m_1 \vec{v}_1 + m_2 \vec{v}_2$$

The x-component of this equation is given by

$$m_1 v_0 = m_1 v_{1x} + m_2 v_{2x}$$

as shown in statement V. The y-component of this equation is given by

$$0 = m_1 v_{1y} + m_2 v_{2y}$$

as shown in statement III. Thus, the answer is B.